

# **Difference in Wind Speeds from Spring to Summer**

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## **Abstract**

The purpose of this paper is to determine whether the end of a season has any impact on wind speed. The seasons we look at are Spring and Summer, so essentially, what we're testing is if there is a significant difference between average wind speeds in June and September. The data used was taken from R, from the package 'datasets' and was collected daily within the same time frame to reduce variability. To determine whether there is any difference in wind speed between seasons, we'll be comparing each month's central tendency and variance because they are main points to consider in this particular analysis.

## **Data Description**

The data was sourced directly from R and is made up of daily measurements of different values of air quality in New York, May to September 1973. There are six variables with 153 data points. The table below shows the names, description, and type of data the variable contains.

| Variable | Type    | Explanation             |
|----------|---------|-------------------------|
| Ozone    | Numeric | Ozone (ppb)             |
| Solar.R  | Numeric | Solar Radiation (lang)  |
| Wind     | Numeric | Wind (mph)              |
| Temp     | Numeric | Temperature (degrees F) |
| Month    | Numeric | Month (1–12)            |
| Day      | Numeric | Day of the month (1–31) |

**Table 1:** Data Description

Some values in "Ozone" and "Solar.R" weren't available, hence labeled 'NA.' In fact, about one fourth of the data points in "Ozone" were 'NA.' The units for some of the groups are as follows:

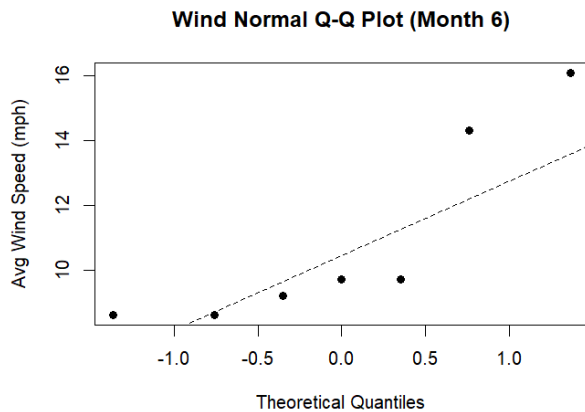
- "Ozone" is measured in parts per billion (ppb) which is a unit of measurement that describes the ratio of a substance's mass to the total mass of a substance. Measured from 1:00 pm – 3:00 pm at Roosevelt Island.
- "Solar.R" is measured in Langley which is typically abbreviated as Ly and essentially measures the amount of solar energy received on a specific area. Measured in the frequency band 4000–7700 Angstroms from 8:00 am to 12:00 pm at Central Park.
- "Wind" is measured in miles per hour (mph) from 7:00 am and 10:00 am at LaGuardia Airport.
- "Temp", is the maximum daily temperature measured in degrees Fahrenheit (F) at LaGuardia Airport.

"Month" and "Day" don't have technical units, rather they go by the standard calendar system with twelve months a year and up to 31 days in the proper month. The months used in the analysis both have 30 days, and only a subset of those days are taken, but more will be discussed in the analysis.

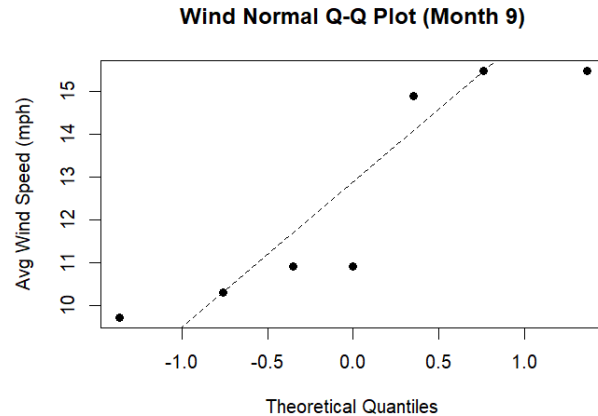
## Analysis

### Preliminary Analysis

My analysis began by looking at details of the variables that were measured. I noticed that while “Wind” and “Temp” has been obtained at the same location, “Ozone” and “Solar.R” were obtained at different locations. Although the data set is about air quality in New York, I would like to have something to compare and ensuring they were obtained in similar areas is important to my particular hypothesis. Thus, I had removed the “Ozone” and “Solar.R” groups from the data since they were not relevant to the analysis. I also removed all months except for June and September and kept only the latter of those months to better reflect the question I’d like to test. Now, between “Wind” and “Temp,” I had tested the normality on multiple different subsets in each group and had decided to run the analysis on “Wind” since that was the only group where a non-normal result was obtained. The plots look pretty clear but it’s good to have multiple confirmations.



**Figure 1:** June Normal Probability Plot



**Figure 2:** September Normal Probability Plot

Thus, I also ran a Shapiro-Wilk normality test. With a p-value = 0.01694, it’s safe to say that the values in month 6, June, is not normal. The same test for the September values resulted in a p-value = 0.0381 which means that both subsets don’t imply a normal distribution and so nonparametric tests are viable. Just to make sure everything is ready to run the analysis on, “Days” were also removed since they essentially act the same as n, being the amount of elements, and “Month” is understood to be the different groups the data is being taken from.

### Main Analysis

I’d first like to test whether the ends of spring and summer have the same average wind speeds; in other words, the first test I’d like to apply is the median test. I’m using the median test because if we can determine whether the population medians are equal, then that means the chance for any given day’s wind speed to exceed the median is the same as the other month. And so, we assume their medians are equal. From now on, I shall refer to month 6 as June and month 9 as September. Using the first letter of the month our naming convention, respectively, the null hypothesis  $H_0$  represents  $M_J = M_S$  and the alternative hypothesis  $H_1$  representing  $M_J \neq M_S$  with  $\alpha = 0.05$ . The next step is to obtain the grand median, and so ordering all fourteen values

from least to greatest lets us see that the grand median is 10.6. Going through the data, there are two values above the grand median in June while there are five in September. Knowing there are seven values in each group, we can make a contingency table holding the amount of values that are above and not above the grand median while keeping track of the totals.

|       | June | September | Total |
|-------|------|-----------|-------|
| Above | 2    | 5         | 7     |
| Not   | 5    | 2         | 7     |
| Total | 7    | 7         | 14    |

**Table 2:** Contingency table for median test

We can see from the table that the speed of wind in June tends to be below the grand median while in September, the speed tends to be higher than the grand median. Before going through with the analysis, this leads me to believe that the time of year, or more specifically, the end of a season has affects wind speeds. Continuing, to find the proportion of data that falls above the grand median, we need to add all 'above' values from June and September and divide by the total size of the data. With  $N = 14$ , we can see that the proportion,  $\hat{p} = (2+5) \div 14 = 0.5$ . Now, plugging each value into the formula for our test statistic, we get  $T = -1.6036$ . This means that our  $p$ -value  $= 2P(Z < -1.60)$  and using the  $Z$ -table, we see that our  $p$ -value  $= 0.11$ . The  $p$ -value obtained from R is 0.1223 which is relatively close to manually obtaining it. That being said, since both  $p$ -values are greater than our chosen  $\alpha$  of 0.05, this indicates that we should accept the null hypothesis at  $\alpha = 0.05$ . This denies the conjecture we created from the table, meaning that the final months of the spring and summer have the same probability for the average wind speed to be above 10.6.

The second test I would like to apply is the Ansari-Bradley test. Knowing whether one month's wind speeds varies more than the other would give more conclusive evidence on if wind speed differs from one season to the next. Also, because there is no significant difference between the medians of wind speed in June and September, the assumption for equal medians is satisfied. Since the medians are equal, the sample with the greater amount of dispersion will receive the smaller ranks.

Now, we begin by assuming equal variance. Thus, the null hypothesis  $H_0$  represents  $\sigma_J = \sigma_S$  and the alternative hypothesis  $H_1$  represents  $\sigma_J \neq \sigma_S$  with  $\alpha = 0.05$ . Going through the sorted data, there are a couple of ties and even one in the center so I took the mid rank of each of them; as for the one in the center, I took the rank to be 6.5 going one way and seven going to other. There are no shared ranks, that is to say, every number has one rank so there shouldn't be an issue in there. The ranks of June are also unaffected by this convention so it also does not affect our test statistic,  $T$ , which is given by the sum of the ranks of June, so  $T = 22$ . Our critical values are  $n_1 = n_2 = 7$  and  $\alpha/2 = 0.025$ . From the table,  $P(T \geq 21) = 0.9744$  and  $P(T \geq 36) = 0.0256$ . Since  $T = 22$  is in between 21 and 36, this implies that we should accept our null hypothesis at  $\alpha = 0.05$ . It's worth noting that if  $\alpha = 0.1$  would also accept  $H_0$ ; however,

any larger than that and our null hypothesis would be rejected since the test statistic barely falls within the bounds for  $\alpha = 0.05$ . Be that as it may, R provided the same value for the test statistic with p-value = 0.1101, which implies that we accept the null hypothesis at  $\alpha = 0.05$ . Seeing the value from R solidifies the fact that a higher alpha would lead us to reject our assumption of equal variance.

### **Conclusion**

The relevant data was fairly clean and the only issue was finding a subset of points that were not normal. That being the case, T-test over the whole interval would be more powerful for similar hypotheses, however median test and Ansari-Bradley test provide similar information. The median test led us to conclude that the wind speeds at the end of spring and summer have the same central tendency and Ansari-Bradley told us that the variance of wind speeds in each month, June and September, are equal. In conclusion, the end of a season has no significant impact on wind speed.